NICOLÒ PAGAN, Ph.D. ETH

AI Research Scientist — M.Sc. Computational Science and Engineering

AI Research Scientist with 10+ years of experience analyzing multi-agent systems, evaluating large language models, and developing ML/AI architectures. Currently leading responsible AI research at University of Zurich, focusing on LLM evaluation frameworks and multi-agent collective behavior. Previously contributed to AGI development at Giotto.AI, working on ARC AGI Challenge. Published 15+ papers including Nature Communications work on game-theoretic network inference and computational Turing Tests for LLM evaluation. Combines strong theoretical foundations in mathematics, algorithms, control theory, and game theory with production-quality Python development skills and proven ability to extract insights from complex structured and unstructured datasets.

Research Focus: LLM — Multi-Agent Systems — Responsible AI — AI Evaluation — Machine Learning — Control Theory — Algorithmic Fairness



20.12.1988, Italian
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WORK EXPERIENCE

Post-doctoral Researcher
Jan 2025 - present

University of Zurich — Social Computing Group — Zurich (CH)

AI Agents: Studying cross-evaluation of multiple AI assistants for real-world deployment
LLM Multi-Agent Systems: Validating generative agent-based models and multimodal LLMs as behavioral social media simulators
Responsible AI: Researching societal and ethical implications of AI multi-agent systems
Research Leadership: Leading 2 PhDs and mentoring 4 Master's students

AI Research Scientist
Mar 2024 - Oct 2024

Giotto.AI — Lausanne (CH), remote

ARC Challenge (AGI): Contributed to gold medal-winning team in the 2024 Kaggle ARC Prize on abstract reasoning; developed computer-vision classification pipeline to reduce solution search space

NCCR Automation Fellow
Jan 2022 - Mar 2024

Swiss National Science Foundation (SNSF) — CH

Responsible AI: Led 432'000 CHF SNSF project on algorithmic fairness and AI ethics in automated decision systems
Feedback Loop Theory: Developed first formal taxonomy of five feedback loop types in ML systems and their mathematical relation to algorithmic biases (EAAMO 2023)
LLM Human Behavior Modeling: Pioneered ChatGPT-based approaches for modeling human behavior in social systems

Lecturer
May 2021 - Dec 2021

University of Zurich — Social Computing Group — Zurich (CH)

HCI: Lecturing Social Computing at the Informatics Department

Doctoral Student
Jan 2016 - Apr 2021

ETH Zurich — Automatic Control Lab — Zurich (CH)

Data Science: Large-scale analysis of fairness in social media recommender sytems
Multi-agent Systems: Modelling the evolution of directed social networks
Game-Theoretic Inference: Developed game-theoretic framework to infer individual behavioral preferences from network structure

Software Engineer and Research Consultant
Sep 2012 - Dec 2015

Ascomp AG — Zurich (CH)

R&D: Development of Computational Fluid-dynamics (CFD) solutions
DevOps: Full-stack software development, HPC, Automated Unit testing

KEY ACHIEVEMENTS

ARC AGI Challenge Gold Medal (2024)

SNSF Grants: 432'000 CHF in competitive funding for Responsible AI research

Nature Communications: Top 50 most-read Physics article (2019)

EDUCATION

Ph.D., ETH Zurich Jan 2016 - Apr 2021
Automatic Control Laboratory

M.Sc., EPF Lausanne Sep 2012 - Apr 2013
Computational Science and Engineering

B.Sc., Politecnico di Torino Sep 2007 - Oct 2010
Mathematical Engineering, cum Laude

AI SKILLS

ML Engineering: PyTorch — TensorFlow — Transformers — LoRA — PEFT
LLM Training: DeepSpeed — TRL — Flash Attention — Accelerate
Programming: Python — C/C++ — Bash — Git — Cloud Computing — HPC
AI Safety & Ethics: Algorithmic Fairness — AI Ethics — AI Alignment

SELECTED PUBLICATIONS

Pagan N., Tornberg P., Bail C.A., Hannak A., and Barrie C., Computational Turing Test Reveals Systematic Differences Between Human and AI Language, *Under Review at PNAS (2026); also presented at COLM 2025 (Conference on Language Modelling) workshop on social media simulations*

Pagan N., Baumann J., Elokda E., De Pasquale G., Bolognani S., and Hannak A., A Classification of Feedback Loops and Their Relation to Biases in Automated Decision-Making Systems, *EAAMO (2023)*

Pagan N., Mei W., Li C., and Dörfler F., A meritocratic network formation model for the rise of social media influencers, *Nature Communications (2021)*

Pagan N. and Dörfler F., Game theoretical inference of human behavior in social networks, *Nature Communications (2019)*